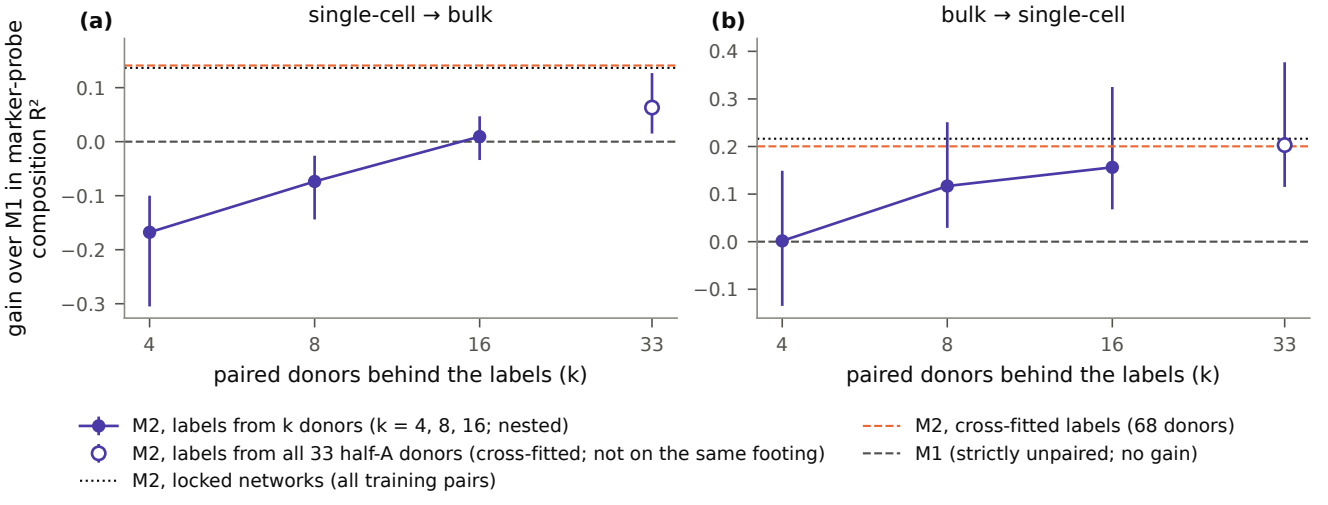
(a)
single-cell → bulk
(b)
bulk → single-cell
gain over M1 in marker-probe composition R²
0.1
0.0
−0.1
−0.2
−0.3
0.4
0.3
0.2
0.1
0.0
−0.1
4
8
16
33
paired donors behind the labels (k)
M2, labels from k donors (k = 4, 8, 16; nested)
M2, labels from all 33 half-A donors (cross-fitted; not on the same footing)
M2, locked networks (all training pairs)
M2, cross-fitted labels (68 donors)
M1 (strictly unpaired; no gain)